

CLAIMS

1. A pharmaceutical and dietary supplement composition having synergistic anti-adipocyte fatty acid-binding protein (aP2), anti-5-Lipoxygenase Activating Protein (FLAP) and anti-Cysteinyl Leukotriene (Cys LT) -1 receptor activities comprising an effective amount of a *Boswellia serrata* extract or fraction selectively enriched in 3-O-acetyl-11-keto- β -boswellic acid (AKBA) and fruit ethanol extracts derived from *Aegle marmelos* for use in the protection from, treatment of and control of inflammatory respiratory disorders, optionally selected from asthma, allergic rhinitis, hay fever, type-1 hypersensitivity and mild allergies, or topical conditions, optionally selected from eczema, psoriasis and wrinkles.
2. A composition for use according to claim 1, wherein the percentage of the enriched *Boswellia* extract(s) or fraction(s) obtained from *Boswellia serrata* in the composition is in the range of 1-95% and the percentage of the extract(s) or fraction(s) of *Aegle marmelos* in the composition is in the range of 5-95%.
3. A composition for use according to claim 1 or claim 2, further comprising an effective amount of extract(s) or fraction(s) or pure isolate(s) derived from *Zingiber officinale* and/or *Garcinia mangostana*.
4. A composition for use according to claim 3, wherein the percentage of the enriched *Boswellia* extract(s) or fraction(s) obtained from *Boswellia serrata* in the composition is in the range of 1-75%, the percentage of the extract(s) or fraction(s) of *Aegle marmelos* in the composition is in the range of 5-75%, the percentage of the extract(s) of *Zingiber officinale* in the composition is in the range of 1- 50% and the percentage of extract(s) of *Garcinia mangostana* in the composition is in the range of 1-50%.
5. A composition for use according to any one of claims 1 to 4, further comprising one or more pharmaceutically, dietically and nutraceutically acceptable additives or excipients.
6. A composition for use according to any one of claims 1 to 5, wherein the enriched extract or fraction of *Boswellia serrata* is an extract standardized to 30% AKBA.

7. A composition for use according to any one of claims 1 to 6, wherein the 3-O-acetyl-11-keto- β -boswellic acid (AKBA) concentration in the enriched extracts or enriched fractions of *Boswellia serrata* varies from 10% to 99%, or from 20% to 60% or from 25% to 45%.

8. A composition for use according to any one of claims 1 to 7, provided in combination with an effective amount of one or more known anti-inflammatory or anti-asthma or immune modulating herbal extracts or fractions.

9. A composition for use according to any one of claims 1 to 8, provided in combination with an effective amount of one or more pharmaceutically or dietetically acceptable anti-oxidants, adaptogens, bioavailability enhancers and trace metals or mixtures thereof.

10. A composition for use according to any one of claims 1 to 9, adapted for administration through oral, parenteral, nasal, inhalation, including nebulizers, transdermal, ocular and any other suitable route.

11. A composition for use according to any one of claims 1 to 10, wherein the composition is in the form of:

a suitable pharmaceutical and dietary supplement dosage form for oral administration, optionally selected from tablets, soft capsules, hard capsules, pills, granules, powders, emulsions, suspensions, sprays, syrups and pellets; or

a suitable pharmaceutical and dietary supplement dosage form for parenteral administration, optionally selected from injections, drops and suppositories; or

a suitable nutritional/dietary supplement dosage form, optionally selected from cereals, baby foods, healthy foods, or food for specified health uses such as solid food like chocolate or nutritional bars, semisolid food like cream or jam, or gel and also beverage and the like, such as refreshing beverage, lactic acid bacteria beverage, drop, candy, chewing gum, chocolate, gummy candy, yoghurt, ice cream, pudding, soft adzuki bean jelly, jelly, cookie and the like.

12. Use of a composition according to any one of claims 1 to 11, wherein a therapeutically effective amount of the composition is administered to humans or animals in need thereof as a pharmaceutical or nutraceutical or dietary supplement for ameliorating conditions related to adipocyte fatty acid-binding protein (aP2) and/or 5 -Lipoxygenase Activating Protein (FLAP) or Cysteinyl Leukotriene (Cys LT) -1 receptor expressions or activities.

13. The use according to claim 12, wherein the amount of the composition or dosage form administered to a human or an animal in need thereof varies in the range of 0.01 to 500 mg/kg body weight/day.

14. The use according to claim 12 or claim 13, wherein the mode of action is through the inhibition of adipocyte fatty acid-binding protein (ap2) expression.

Zingiber officinale (Ginger) has been used as a treatment for nausea, diarrhea and epigastric and joint pains in Ayurvedic and traditional Chinese medicine. Ginger has antihistaminic activity. Ginger's oleoresin constituents have potent inhibitory effect on the biosynthesis of pro inflammatory prostaglandins and leukotrienes. Shogaol and Paradol inhibit COX-2 activity and Gingerol inhibits 5-LOX enzyme activity (Tjendraputra, E., et al., *Bioorg. Chem.*, 29, 156-163, 2001; Srivastava, K. C., et al., *Med. Hypoth.*, 39, 342-348, 1992). In an uncontrolled trial, all seven arthritis patients treated with half gram of powdered ginger per day reported pain relief and decreased signs of inflammation (Srivastava, K. C., et al., *Med. Hypoth.*, 39, 342-348, 1992). Ginger root extracts standardized to 5 - 15% gingerols are used to prepare this composition. Extracts standardized to other concentrations and other components could also be used for making the present inventive compositions.

Garcinia mangostana L belongs to Guttiferae family. It is commonly known as Mangosteen. Mangosteen is a slow-growing tropical, evergreen tree and can attain 6 -25 m in height with leathery glabrous leaves (B. N. Sastry, *The Wealth of India Raw materials*, Vol. IV: pp 103-105, 1956). The tree is mainly found in India, Myanmar, Sri Lanka and Thailand. The edible fruits of this plant are considered to be one of the best of all tropical fruits. In the ayurvedic system of medicine, the fruit hull of the plant finds wide application, mainly as an anti-inflammatory agent and in the treatment of diarrhea (V. Peres, et. al., *Phytochemistry* 55: pp 683-710, 2000; V. Peres et. al., *Phytochemistry*. 44: pp 191-214, 1997). *Garcinia mangostana* alcohol extract or hydroalcohol standardized to α -mangostin is used in the present invention. However *Garcinia mangostana* extracts selectively enriched in γ -mangostin may also be used.

The use of *Boswellia* extract is known as a dietary supplement to control inflammatory conditions. For example, US2006/040000 discloses a composition comprising a mixture of *Boswellia* extract, salts of glucosamine and curcuminoids. The composition also optionally contains bromelain, chondroitin, methylsulfonylmethane, resveratrol, extracts of white willow and ginger and quercetin.

Aegle marmelos extract is also known to treat certain medical conditions. For example, Arul, V et al. (*fighter medicine*, 11, 679-683, 2004) discloses the use of the alcoholic extract of the leaves of *Aegle marmelos* to treat asthma. Lankapatiravana (Indradeva Tripathi, 1995, Krishnadas Academy, Varanasis, India, page 117) discloses a formulation containing extracts of both *Aegle marmelos* and *Zingiber officinale* along with a few other ingredients used for the treatment of allergenic rhinitis and vasomotor rhinorrhoea. Panditarao (1990, Govt of India: Sahastrayoga, New Delhi, page 63) discloses another formulation containing extracts of both *Aegle marmelos* and *Zingiber officinale* along with a few other ingredients used for the treatment of bronchial asthma through oral administration.

Objects of the invention:

The major objective of the present invention is to provide Pharmaceutical/Dietary supplement compositions comprising *Boswellia serrata* extract selectively enriched in 3-O-acetyl-11-keto- β -boswellic acid (AKBA) and extracts of *Aegle marmelos* for improved therapeutic effect in the prevention, treatment and control of asthma and other inflammatory diseases.